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# Granular flow and segregation in a four-bladed mixer

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**Abstract**

In this study, we experimentally examine flow and segregation of granular material in a cylindrical mixer geometry agitated by four 45° pitched blades, which is representative of equipment such as high-shear granulators and filter-dryers. We observe that the free surface of the granular bed deforms, rising where the blades are present and falling between blades passes. Using particle image velocimetry (PIV), we measure the instantaneous, average, and fluctuating velocity fields at exposed surfaces (top surface and near the wall), for both near-monodisperse and polydisperse granular materials. The radial and axial point-velocity profiles indicate three-dimensional recirculation patterns indicative of avalanching and bed penetration. For polydisperse mixtures, we find that depending on the shear rate, different segregation mechanisms can take place. Under low shear, complex lobe and striation segregation patterns occur through stretching and folding due to surface avalanching. This leads to enhanced initial mixing rates in a manner consistent with spontaneous chaotic granular mixing. At high-shear rates, segregation is controlled by the rotation of the blades. As a result, coarse particles have a tendency to migrate both to the free surface and the outer wall independently of initial bed loading conditions.
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## 1. Introduction

Agitated granular beds are common in a variety of particle processing technologies. While in many cases, agitated mixers are viewed as a mean to homogenize a blend of solid particles, they also serve to enhance chemical reactions or to improve heat and mass transfer processes. Such equipment is also used for granulation and to encourage flow, as in agitated hopper and feed-frame assemblies in tablet presses. Thus, quality and consistency of products ranging from commodity chemicals to consumer goods to pharmaceutical formulations depend on reliable characterization of granular mixing. For example, Lekhal et al. (2003, 2004) showed that for agitated drying of crystals the final particle size is a function of attrition and agglomeration which are very sensitive to the flow patterns within the particle bed. Despite their extensive use, the flow of granular materials in agitated devices and the mechanisms by which particle motion is generated are still poorly understood. Even less is known about associated segregation patterns due to differences in particle properties, and the impact of process parameters and material properties on the laboratory or industrial scale. This can result in poor process control, and complicates scale-up. Many investigations of blade motion through granular solids in agitated beds have concentrated on power and torque requirements (Bagster and Bridgwater, 1967). The aim of this paper is to provide some basic features of solid flow patterns in mechanically agitated granular beds and to examine associated size segregation in polydisperse particle mixtures.

This work concerns the dynamics of particles shearing by boundaries in the presence of gravity. Unlike molecular fluids, these dynamics are not yet described by approaches analogous to the Navier–Stokes equations, as materials composed of discrete particles are characterized by more complex, dissipative rheologies. They exhibit a tremendous range of behaviors, from solid-like, quasi-static flows

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to rapid, fluid-like flows (Jaeger et al., 1996; Tardos et al., 2003). Early studies on mixing of granular materials in cylindrical bladed mixers were conducted by Novosad (1968), Bagster and Bridgwater (1967, 1969), Bridgwater et al. (1968) and Malhotra et al. (1988, 1990). While Bagster and Bridgwater (1969) and Bridgwater et al. (1968) essentially focused their analysis on particle displacement in horizontal beds only, Malhotra et al. (1988, 1990) also examined the velocity distribution and flow patterns. In most of these studies, the authors confirmed, through observation of the surface, the formation of a heap of particles moving just ahead of the blade, and the presence of a wake region with high bed porosity behind it. It was also found that there is no vertical motion in the clearance between the blade and the wall (Bridgwater et al., 1968; Malhotra et al., 1988). This can be crucial for processes where heat transfer is taking place through the walls as is the case in agitated contact drying. In recent years, the characterization of granular flow in vertical vessels was drastically improved by 3D visualization of the granular bed using positron emission particle tracking (PEPT) (Stewart et al., 2001a,b) and by numerical simulations using the discrete element method (DEM) (Stewart et al., 2001b; Zhou et al., 2004). Both methods allow the description of internal flow structures within the granular bed with a great deal of accuracy. Stewart et al. (2001a) showed through PEPT measurements in a cylindrical mixer equipped with two opposing flat blades that at the wall the particles in the path of the blades are displaced upward forming a heap and then moved over the blades. Away from the wall, a fraction of the upwardly displaced material moved away from the blade down the faces of the heaps. Another fraction was seen to move radially, which revealed 3D structures within the granular bed. The experimental results obtained by PEPT and DEM simulations of Stewart et al. (2001b) agreed well. Color coding particles in DEM simulations allowed Zhou et al. (2004) to better analyze the mixing mechanisms within agitated granular beds. In particular, they found that experimentally observed flow patterns within the agitated bed can be matched by adjusting particle–particle and particle–wall sliding and rolling friction coefficients.

While considerable progress in mixer design has been achieved, characterization of mixing-segregation transitions in such geometries is scant. This is somewhat surprising given that segregation of components is a leading cause of production failures due to non-uniformity, and that particle maldistribution often negatively affects the efficiency of granulation or drying processes. In all the studies cited above, the results were interpreted and analyzed with respect to mixer geometry (vessel dimensions, and blade height, number, and pitch) and process parameters (such as fill level and agitation speed). Relatively little attention has been paid to the effect of particle size, shape, or density distributions in polydisperse materials on flow patterns. Multiple components and particle size distributions clearly add to the difficulties in predicting flow patterns of granular materials, and particle mixing, via convection or diffusion mechanisms, and segregation are intimately linked. No general way yet exists to predict a priori, if two powders of different characteristics will mix or segregate (in the lab or upon scale-up) (Shinbrot and Muzzio, 2000).

Most studies of segregation in mixers have focused on low shear, diffusive tumbling blenders, that have been shown particularly susceptible to segregation (Shinbrot and Muzzio, 2000; Ottino and Khakhar, 2000; Alexander et al., 2003). In contrast, cylindrical mixers with impellers are usually assumed to mix via convection to small scales of scrutiny, albeit with greater stresses imposed on particles (Poux et al., 1991). Studies in high-shear granulators (e.g. Plank et al., 2003) have addressed flow patterns for materials possessing wide particle size distributions, but have not specifically addressed associated segregation. To the best of our knowledge, only the numerical study of Zhou et al. based on DEM (Zhou et al., 2003) has specifically examined the effect of size and density segregation on granular mixing within cylindrical agitated vessels. As with monodisperse cases, particles in front of the blade rose to form a heap and then flowed downward on the bed surface over the blade or to the base of the heap to rejoin the flow toward the blade. Zhou et al. (2003) also quantified segregation of large and light particles to the top layer of the mixing bed, while small and heavy particles remained at the bottom and in front of the blades. The segregation was not explained rigorously using either heuristic or theoretical arguments (based on gradients in density or fluctuational velocities) that have been applied in other mixer geometries (Ottino and Khakhar, 2000; Alexander et al., 2003; Khakhar and Ottino, 1999).

The main goal of this study is to analyze the mixing behavior of dry granular materials in a vertical cylinder equipped with a four bladed mixer using the particle image velocimetry (PIV) technique and visual observations. This geometry is representative of several granular processes including filter dryers and high-shear granulators. PIV allows the characterization of the flow field (instantaneous velocities, average velocities and velocity fluctuation) at the free surface and near the transparent wall, while image analysis is used to determine particle trajectories and quantify mixing and segregation kinetics. Although, the characterization of the flow pattern at exposed surfaces (free surface and walls) does not allow a full understanding of the mixing behavior of the entire granular bed, it remains crucial for several granular processes. For example it is important in granulation to determine the flow pattern of the top layer of the bed in order to optimize the addition of the moisture (rate, location, etc.). In addition, understanding the flow and mixing at the surface is a first step towards developing an understanding of the full three-dimensional (3D) behavior of the bed. First, we determine particle trajectories and velocity fields at exposed surfaces of the agitated bed for monodisperse materials. Next, we describe observed segregation patterns for polydisperse materials, which we then relate to the magnitude of measured granular temperature gradients using granular kinetic theory at high densities (Khakhar and Ottino, 1999). Our

findings are anticipated to have immediate practical applications for sampling and process optimization. The observed segregation patterns have implications for homogeneity control, and may suggest critical locations suitable for sampling points, as well as process modifications to maintain uniformity. The paper is organized as follows. In Section 2 we describe the experimental set up and PIV technique. In Section 3 we present results on monodisperse and polydisperse materials and we conclude in Section 4.

## 2. Experimental setup

### 2.1. Experimental apparatus

Our bladed mixer geometry is shown in Fig. 1(a). The upper surface of the bed is unconfined, and is free to deform due to action of the agitator blades. This geometry effectively reduces to a combination of bounded shear and gravity-driven flow on inclined planes (oriented at right angles to one another). The geometry differs from other studies: we use a *pitched* blade agitator typical of industrial applications. Four 45° pitched blades, equally spaced around the agitator shaft, impart azimuthal, axial and radial velocities to particles in the bed. The direction of rotation with respect to the granular bed is termed "acute" or "obtuse" as shown in Figs. 1(b) and (c), respectively. Our focus is on the obtuse blade orientation most common in particulate processing equipment. However, when significant differences are obtained using the acute geometry, these are highlighted, as these results relate to certain operations in which blade

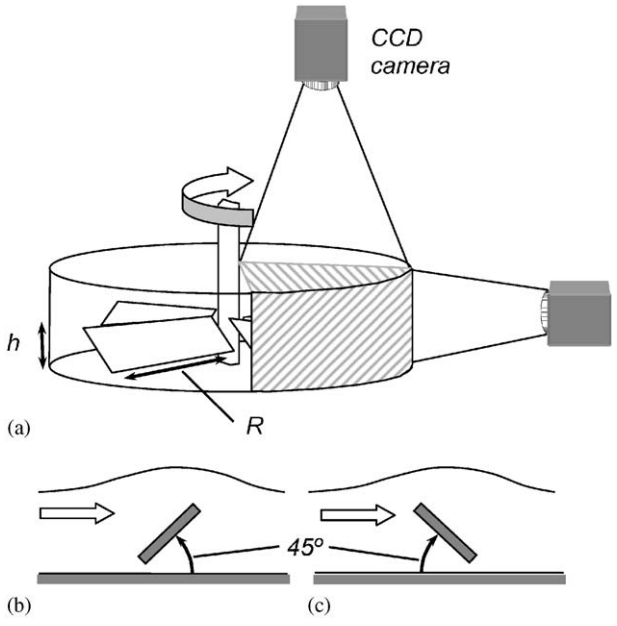



Fig. 1. (a) Schematic of 4-blade agitated bed geometry. Crosshatched regions interrogated by PIV. Pitched blades viewed end-on, through outer wall in (b) 'acute', and (c) 'obtuse' orientations. Arrows show direction of blade movement.

pitch (or the direction of rotation) is changed, such as during the unloading step for agitated filter-dryers.

Granular material is loaded in the 1 L cylindrical vessel to an initial depth of 25 mm. This is sufficient to just reach the top of the agitator blades, which are constructed of polyethylene and screwed to a mild steel shaft. The agitator can be driven forwards or in reverse (providing the obtuse, and acute blade pitches, respectively) using a speed controller with accuracy of $\pm 0.1$ rpm under load. There is $\sim$1–2 mm clearance between the bottom of the blades and the base of the container, and a 5 mm gap between the outer edges of the blades and the vertical vessel walls. Other fill volumes have been examined, but we find the velocity and segregation trends we report for a 25 mm initial depth are representative of other depths, although quantitative variations occur.

### 2.2. PIV measurements

While improved characterization of flow in agitated granular beds is recognized to be a requirement of uniformity control, the traditional approach consists of withdrawing samples for offline analysis and deriving statistical functions to describe the system (Poux et al., 1991). This is also the dominant approach still used for quality control during production. More detailed characterization of flow patterns is also possible by bed freezing (via solvent infiltration and setting) and dissection (Shinbrot et al., 1999a,b; Wightman et al., 1996), although sampling and bed freezing procedures are intrusive methods which are very laborious and subject to some uncertainty. Recent studies on particle mixing are based on newly advanced non-invasive techniques. Malhotra et al. (1988, 1990) viewed a cylindrical stirred tank through one end wall and interpreted their findings in relation to particle motion in the vicinity of the blade. Another alternative is the PIV technique—widely used to characterize fluid flows—which can be used to track individual particle motions in a granular bed from high-speed CCD images. Analysis of various materials in different vessel geometries is relatively straightforward to implement, and yields both instantaneous and mean velocity vector fields, at maximum rates of several thousand hertz. This, together with a high spatial resolution has permitted PIV to be used in several archetypal granular geometries (Goldfarb et al., 2002; Conway et al., 2003). The chief disadvantage of this technique for opaque materials is that only 2D flow information is typically accessible, either at the bed free surface, or through transparent sidewalls. 3D visualization of particle motion can be obtained using MRI technique (Metcalfe et al., 1998) or the PEPT (Parker et al., 1993; Bridgwater et al., 1993), in which a radioactive tracer particle is tracked by positron-sensitive detectors as it moves within the bed. However, information collected from the PIV technique can be used to develop an initial understanding of the mixing behavior of granular material in the agitated mixer. This can be viewed

as a first step towards developing numerical tools, which can be used in a systematic way to simulate the motion of ten of thousand of particles. The simulations can be used to infer the bulk behavior of the granular bed as well as other quantities that are not easily accessible experimentally (e.g. force and stress distribution within the bed).

In our experiments, a CCD camera with a spatial resolution of 0.15 mm/pixel was placed directly above one quadrant of the flow and used to record images of the free-surface at 500 frames per second. The image spans the entire radius of the container. A second camera location enables images of flow around the blades visible through transparent sidewalls to be obtained. Sequential images are processed using cross-correlation and a $3 \times 3$ interrogation cell smoothing filter (Dantec, Inc). Mean velocity fields are obtained by averaging over a minimum of 500 frames. Additionally, a 30 fps digital video camera enables high resolution images of the entire apparatus to be obtained under controlled lighting, for subsequent image analysis.

## 3. Results

### 3.1. Flow characterization for near-monodisperse materials

Before analyzing segregation patterns for unlike components, we characterize the mean and fluctuating velocity fields for near-monodisperse materials. Dry art sand ($355 < d < 425\,\mu m$) containing both light and dark grains aids visualization and PIV analysis (the different colored grains are mechanically identical). The blades are agitated at 3–300 rpm using the obtuse blade pitch, which, for a blade length $R$ of 45 mm and bed depth of 25 mm, corresponds to average dimensionless shear rates $\gamma^{o*} = \gamma[d_p/g]^{1/2}$ of 0.06 to 5.7. $\gamma$ is the shear rate, $d_p$ the bed depth and $g$ is the gravitational acceleration. This value of $\gamma^{o*}$ spans the entire range of granular flow regimes, from slow, through intermediate to rapid granular flows, as defined by Tardos et al. (2003). Truly rapid flows ($\gamma^{o*} > 3$) are calculated to occur for rotation rates above $\sim 160$ rpm in this geometry, while the slow regime ($\gamma^{o*} < 0.15$–$0.25$) here correspond to rotation rates below 0.6 rpm. After initiating agitation, we perceive no change in bed features after a few seconds of blade rotation, but nevertheless wait at least 2 min before obtaining free-surface or side view velocity images.

The instantaneous velocity fields obtained from PIV permit mean and fluctuational velocity information to be computed. The latter is commonly termed granular "temperature" and is defined as $T = 1/2\langle \boldsymbol{u}'\boldsymbol{u}' \rangle$, where $\boldsymbol{u}'$ is the fluctuating velocity computed relative to average velocities in the spatial location (pixel) being examined, while $\langle\,\rangle$ represents temporal averaging for the pixel in question. Associated temperature gradients have been shown to induce segregation in particle mixtures (Khakhar and Ottino, 1999; Luding and Herrmann, 1999; Dahl and Hrenya, 2004).

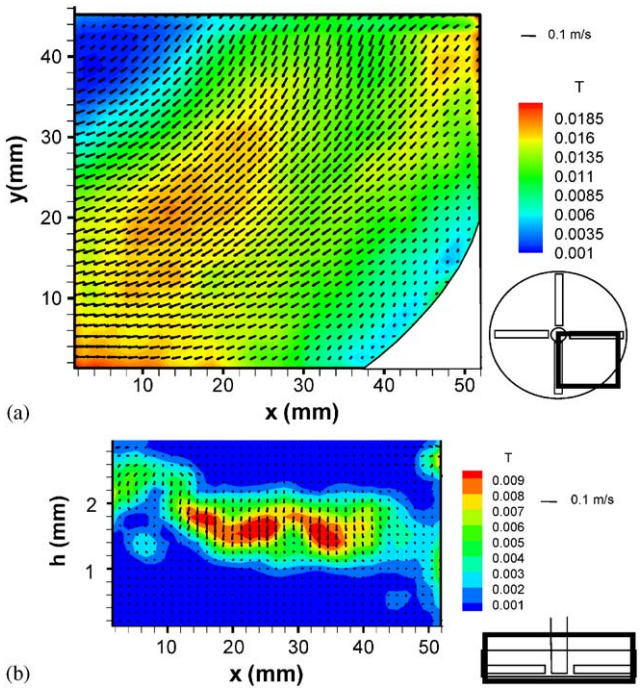



Fig. 2. Mean velocity vectors and granular temperature fields for obtuse agitator pitch at 50 rpm (clockwise). Schematics depict image area. (a) Overhead view of single quadrant. (b) Side view of entire bed. Fields obtained by PIV from cross-correlation of 500 frames, taken 2 ms apart.

Mean velocity and granular temperature fields for particles on the surface and at the walls for our current geometry are shown in Figs. 2(a) and (b), respectively. Measurements for 50 rpm are depicted, obtained from 500 sequential image frames (i.e., 1 s real-time flow data), and this encompasses several blade passes. Mean velocities are predominantly azimuthal. Granular temperature minima at both the outer edge of the flow and close to the agitator shaft indicate that surface fluctuations are greatest far from radial boundaries. The side view shown in Fig. 2(b) reveals that the bulk of particles near the wall possess low velocity and temperature. Low velocities (relative to the free surface) may be due to the existence of a 2 mm clearance between the blade tip and the wall, in which shear planes exist (Nedderman and Laohukal, 1980) and hinder transmission of blade motion to the walls. Temperatures measured via side view images are small indicating that the bed near the wall is uniformly "cool", likely a result of wall friction. Due to the motion of the agitator, this is not expected to be the case in the bed interior.

We use the instantaneous data provided by PIV to characterize transient surface features created by blade passes. For the obtuse blade pitch, the free surface deforms; rising where blades are present, and falling between blades passes. In Fig. 3(a) a CCD image of surface of flow is shown in which a blade is captured in the diagonal position spanning the top left to the bottom right of the image. At 50 rpm, a crest forms directly above blade position. Corresponding

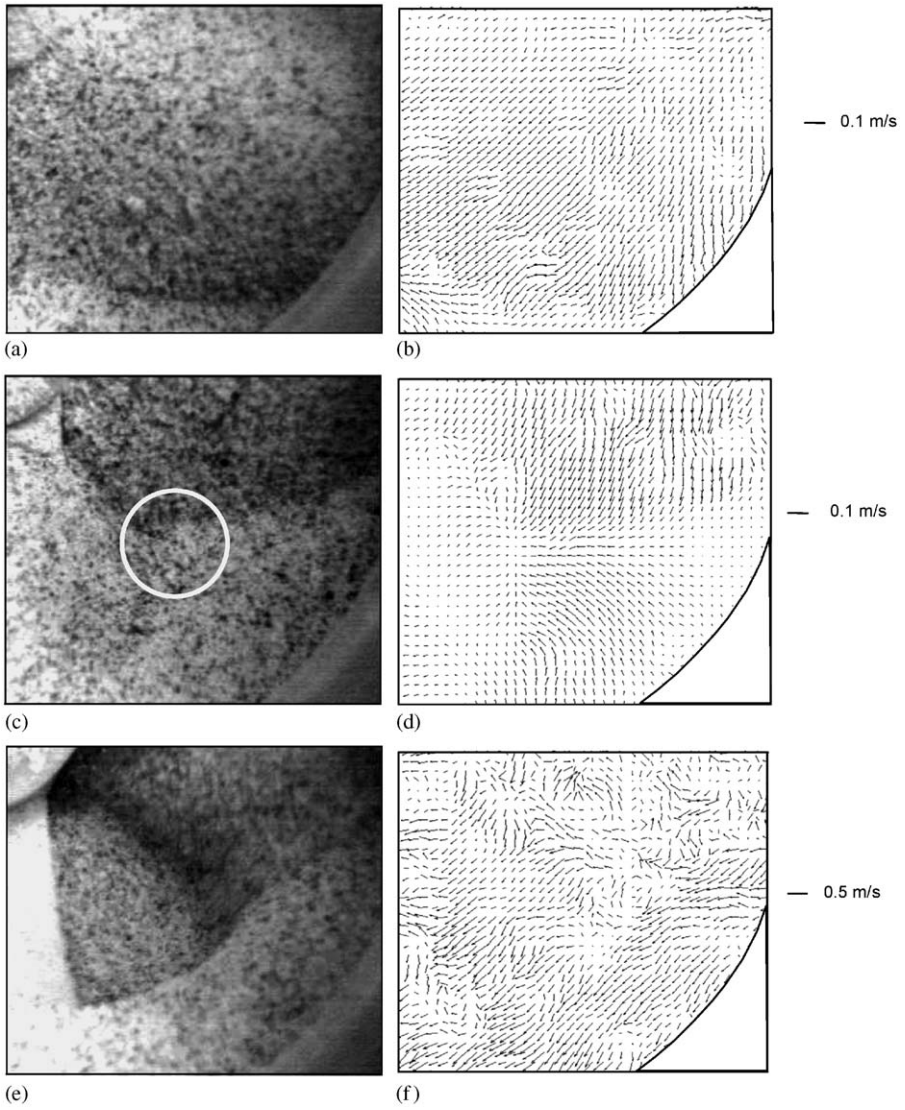



Fig. 3. Instantaneous overhead views of single quadrant with obtuse agitator pitch. Blade rotation is clockwise. (a) At 50 rpm, image of 'peak', with blade captured in the diagonal position (b) corresponding PIV velocity vectors. (c) At 50 rpm, image of 'valley' with blade captured in the square position. (d) Corresponding PIV velocity vectors indicating avalanching secondary flow. (e) At 300 rpm, image of 'valley', with blade captured in the square position captures significant "cavitation", and (f) corresponding PIV velocity vectors, indicating significant circulation. Circled area represents sink point.

instantaneous velocity vectors are shown in Fig. 3(b). For the most part, instantaneous vectors reflect the mean azimuthal trajectories of Fig. 2(a). In contrast, as the blade moves out of the image frame, a valley is created in the absence of underlying blade support, and this can be discerned in Fig. 3(c). The depression is comprised of three distinct sloped surfaces converging near the center of the image (circled in Fig. 3c). Furthermore, instantaneous velocities in Fig. 3(d) show that particle motion is downward on these slopes, indicating that avalanching occurs.

Such secondary flow patterns at the surface are not discerned from mean velocity information (Fig. 2), but the flows clearly contribute to substantial velocity transients as each blade passes the interrogation area, and suggest that associated subsurface circulation patterns exist. Similar circulations patterns were described as material avalanches both ahead of and behind flat blades, coupled with subsurface replenishment flows (Stewart et al., 2001b; Zhou et al., 2004). Here, this effect also occurs, but the proximity of each blade to the next causes forward-flowing avalanches from one blade to merge with backward-flowing avalanches from the blade immediately upstream, resulting in the sink points captured in Fig. 3(c). The third avalanche evidently occurs through interactions with the wall, as material accumulated there flows inwards (Poux et al., 1991). The vertex of the three surfaces thus appears to act as a material sink—a point at which particles collect with no accumulation of the granular material. This strongly suggests that the

particles are moving downward towards the bottom of the bed, which reveals the existence of 3D structure within the bed. Similar behavior has been observed by Stewart et al. (2001b) using PEPT technique and DEM simulations for a two bladed impeller. Although, the PIV measurements offer only the possibility to directly access the surface behavior of the granular bed, it is possible to make some inferences with regard to the complex behavior of the solid particles within the bed.

PIV permits accurate velocity measurements at drastically increased rotation rates. For example, at 300 rpm, our blade tip speed is calculated to be 6.4 m/s, resulting in maximum particle translations that are still within the 40 mm imaging frame. This alters velocity gradients and associated circulations, as shown in Fig. 3. At 300 rpm, the high-speed imaging reveals that the valleys are now almost devoid of particles, creating a wedge shaped cavity behind the passing blade. This is accompanied by significant velocity differences between particles flowing rapidly at the edge of the void and the void itself. Voids grow with blade speed, and are reminiscent of bubbles formed behind impellors in cavitating liquids.

Periodicity due to blade passes and signs of extensive recirculation are revealed as we examine point velocities over time. Fig. 4 shows velocity time profiles taken near the center of the PIV images in Fig. 3 (i.e., midway between the shaft and outer wall). In Figs. 4(a) and (b), we show both radial and azimuthal velocities profiles for several rotation rates. At 3 rpm, radial velocities are insignificant and particles move almost exclusively azimuthally in the direction of blade rotation. Higher rotation rates allow several blade passes to be captured, and at 50 rpm, we see that both radial and azimuthal velocities fluctuate with wavelength of $\sim$300 ms, approximating the 270 ms period expected for subsequent blade passes at this speed. Radial motions towards the wall (i.e., positive values) occur at times in which the azimuthal motion is in the direction of blade rotation. During spells of outwards radial motion, azimuthal velocities are briefly reverse. These point velocity profiles strongly suggest that particles recirculate radially on the free surface, at times moving in the opposite direction to the blades. The magnitude of surface circulation does not vary linearly with rotation rate. Negative azimuthal flows become insignificant at 100 rpm, but emerge once more above 200 rpm.

Understanding of the curious suppression and re-emergence of reverse particle flows is provided by examination of the high-speed video footage. At 50 rpm, we observe particles avalanching down the leeward slopes behind the blades, falling faster under gravity than the speed of the blades, permitting flow recirculation. However, higher agitation causes the crest surrounding blades to move more quickly than particles fall in the opposite direction. Reverse flow ceases. By contrast, above 200 rpm wedge-shaped voids emerge, (Fig. 3e), and it appears that rapid back-filling of the voids by material accumulated at the outer wall encourages some reverse particle motion (c.f. rapid collapse of vapor bubbles in liquid cavitation).

Velocity–time profiles also vary with radial locations (Fig. 5). Both periodicity and amplitudes show little evidence of the blade frequency ($\sim$300 ms) next to the shaft, but strong radial oscillations and negative azimuthal velocities 10 mm can be observed from the shaft. Circulation flows appear to dampen closer to the outer wall. Furthermore, we note the fine structure of the profiles, which indicates a second group of fluctuations with frequencies exceeding those of blade rotations becoming significant at radial positions far from the shaft. Again, the source of these high-frequency fluctuations is suggested by examination of high-speed video footage. We observe high-frequency surface avalanches, which become larger and more frequent closer to the outer wall.

### 3.2. Flow characterization for segregating materials

Near-monodisperse flows reveal several features known to cause mixture segregation: circulation patterns, granular temperature gradients, and occurrence of avalanches at several frequencies (Luding and Herrmann, 1999; Knight et al., 1993; Makse et al., 1997). Replacement of art sand in the agitated vessel with mixtures of glass particles of varying diameter allows size segregation to be examined. Mixtures are initially well mixed by agitation of small aliquots prior to loading, but it is found that the segregation patterns obtained are insensitive to loading conditions. Within 10 revolutions of the agitator blades, the mixtures reach a state of segregation that subsequently appears time invariant. Nonetheless, we again waited for at least 2 min before recording images or obtaining velocity data. We find that the observed segregation follows two distinct patterns, depending on shear rate, and we now describe each in turn.

#### 3.2.1. High-shear operation

We define high-shear flows as those produced by agitation above 50 rpm, i.e., $\gamma^{o^*} > 0.95$, which, as we find, share many traits with the rapid flow regimes in which $\gamma^{o^*} > 3$ (Tardos et al., 2003). In Figs. 6(a) and (b) we show significant segregation in overhead and side views of a mixture of three particles sizes agitated at 50 rpm (see caption for details). Coarse, red particles segregate from finer, white particles radially, to the outer wall, and vertically, to the free surface. The finest (black) particles also segregate downwards to disperse groupings near the bed base. The dominant process is the rise of the large diameter particles, as we observe this to occur readily at all agitator rotation rates (we find radial segregation to the outer wall does not occur at lower shear, $\sim$15 rpm). Similar segregation patterns are observed for particles an order of magnitude smaller in size, as shown in Figs. 6(c) and (d) in which large, rust colored particles migrate within a bed of small, white particles to both the outer walls and to the free surface.

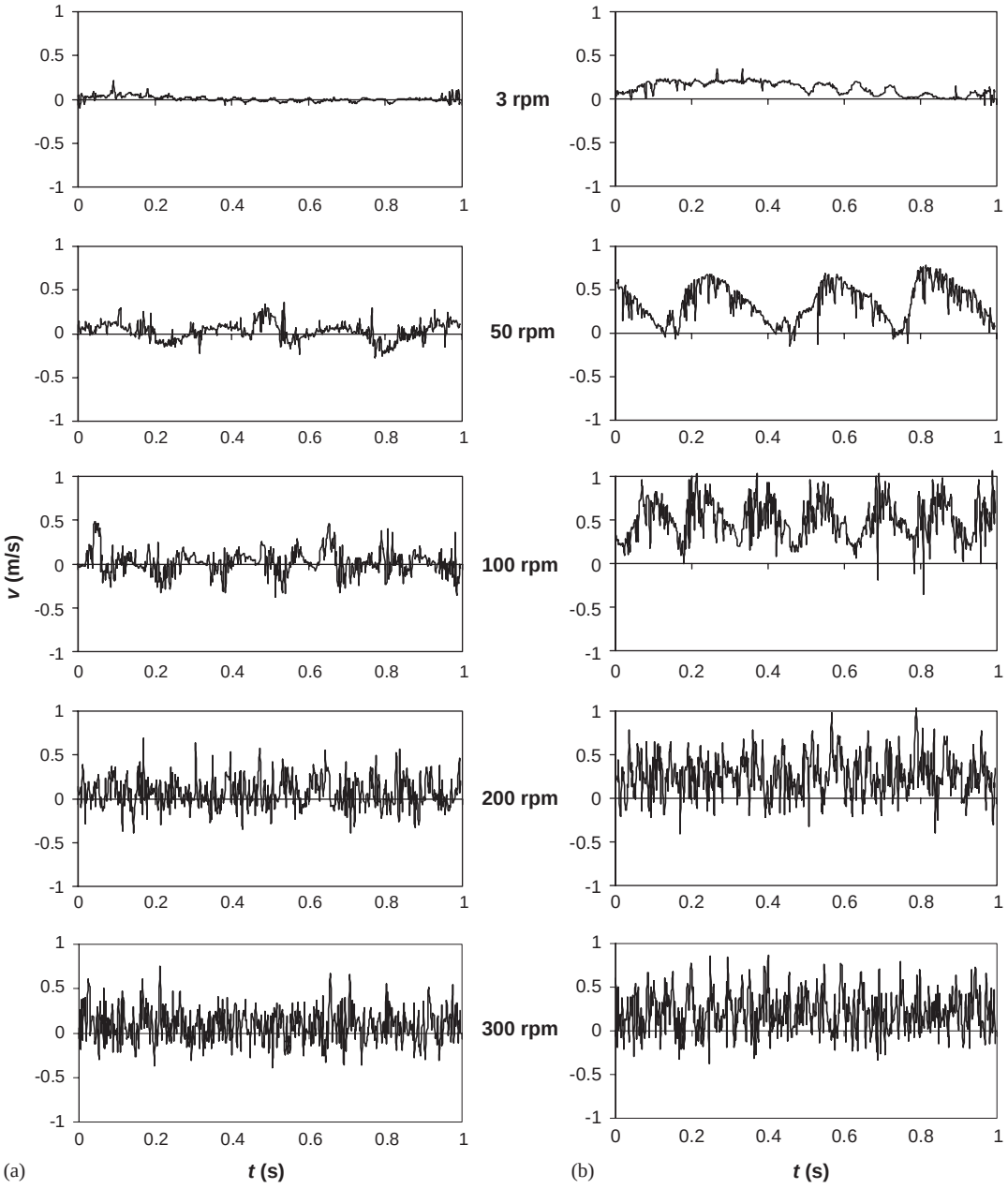



Fig. 4. Effect of increasing rotation rate. From top views of single quadrant (a) radial and (b) azimuthal particle velocities from PIV, taken at center of blade (20 mm from shaft), plotted against time for 3, 50, 100, 200, and 300 rpm.

Observed segregation patterns are consistent with earlier discrete simulations of Zhou et al. (2003), and the well known mechanism of sieving segregation, in which small particles enter inter-particle spaces created in shear zones, forcing less mobile large particles to rise. Recall that Fig. 2(b) (side view) indicates higher granular temperatures measurements near the free surface than for submerged material. Particles in the granular bed are clearly closely packed, and under these dense conditions kinetic theory predicts that small particles will migrate to the low-temperature regions near the bed base (Khakhar and Ottino, 1999). Thus, vertical segregation of large particles to high-temperature regions is expected, which is consistent with the sieving mechanism. A different situation arises on the free surface where particles are not constrained by overlying particles and effectively experience more rapid, dilute conditions. Granular temperature is lowest near the outer wall (Fig. 2a) and so thermal diffusion flux of large particles to this wall can occur. The necessity of relatively high rotation rates in producing radial segregation (large particles cover the free surface uniformly below 15 rpm, not shown) can be interpreted in two ways. First, granular temperature gradients

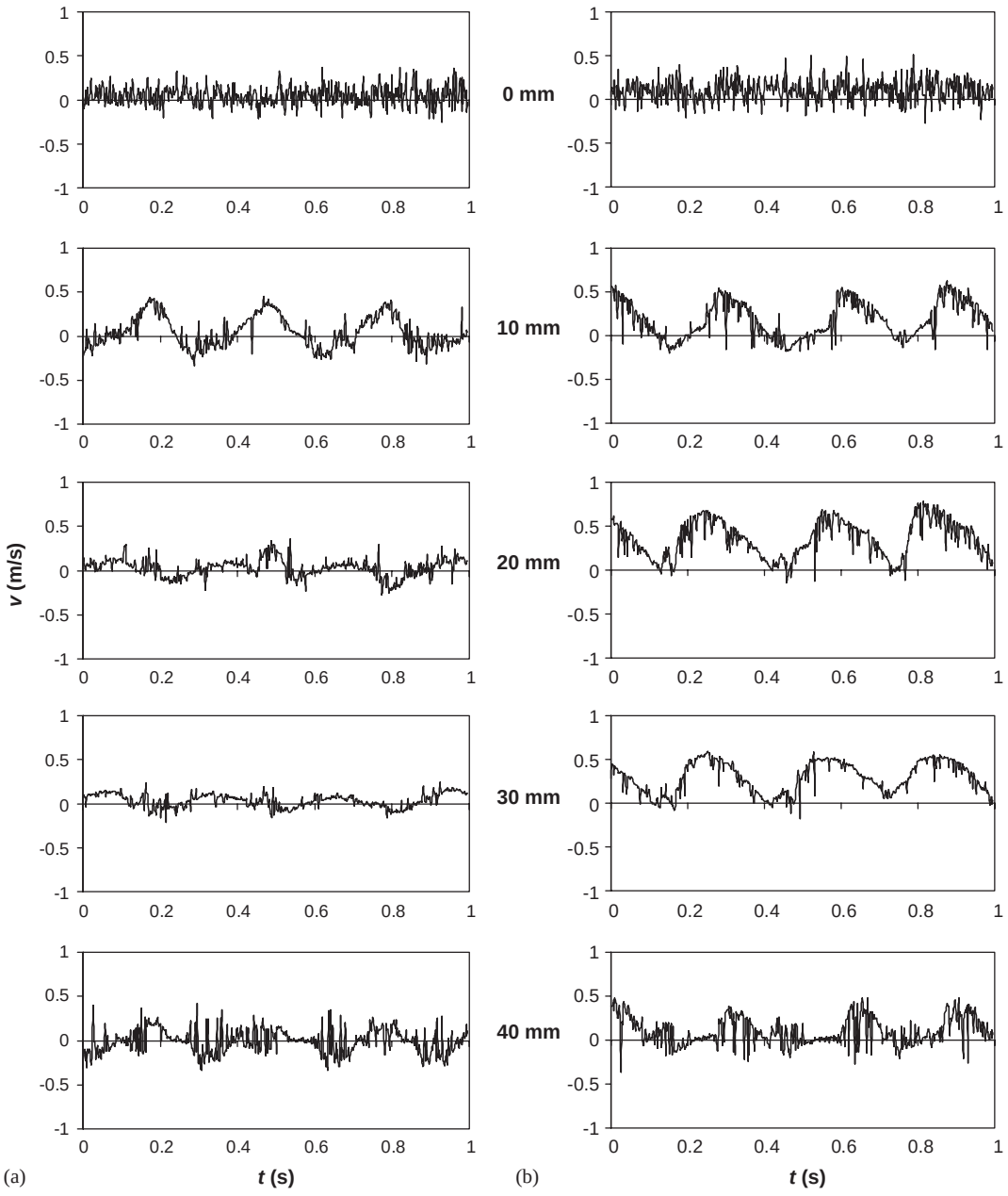



Fig. 5. Effect of increasing radial position for fixed rotation rate of 50 rpm. From top views of single quadrant (a) radial and (b) azimuthal particle velocities from PIV, taken at radial positions 0, 10, 20, 30, and 40 mm (measured from shaft).

increase with the blade rotation rate, such that the driver for the thermal diffusion flux diminishes below a certain agitation speed. Secondly, the data are consistent with another heuristic mechanism caused by the ability of large particles to flow over a rough granular surface more readily than smaller particles. In our case, centrifugal force plays the role of gravity in effecting this form of segregation, driving large particles rolling preferentially to the outer wall.

### 3.2.2. Low-shear operation

We observe a new and dramatic change in segregation patterns when agitation is slowed below 10 rpm, corresponding to intermediate and quasi-static flows with $\gamma^{o^*} < 0.2$. Fig. 7(a) is an overhead view of the entire bed free-surface under high shear as examined in the proceeding section exhibiting radial and vertical segregation described above. By contrast, Fig. 7(b) shows the same bed at 1 rpm. Segregation of an entirely different character is observed. Large particles not only congregate around the circumference at the free surface, but form numerous striations projecting radially from the outer walls and curved regions at the edge of crests over blades. In Fig. 7(c), large particle accumulation at the free surface is evident. However, in this case the large particles overlay a thin white layer of white layer comprised of small

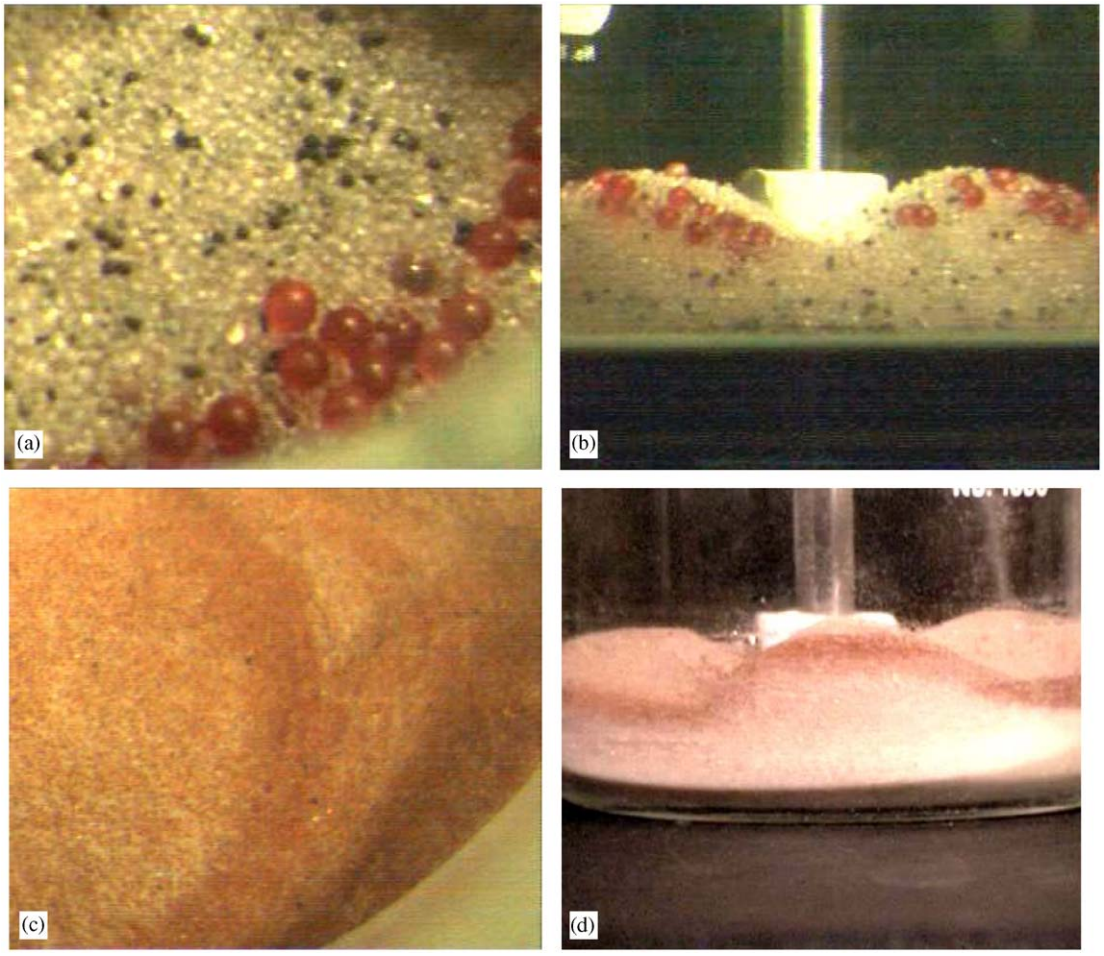



Fig. 6. Observed segregation patterns. Mixture of 0.9, 1.6 and 4 mm glass spheres agitated at 50 rpm (counterclockwise) with obtuse blade pitch shown in (a) overhead view (b) side view. Mixture of 125 and 450 μm glass spheres agitated at 50 rpm with obtuse blade pitch shown in (c) overhead view (d) side view.

particles. This in turn overlays a more mixed region in the bulk. Close-up surface images are shown in Figs. 7(d)–(g) as a function of rotation rate. Under high shear (Fig. 7d) the only significant feature is large particle segregation to the walls, although a faint, diffuse band appears to outline an avalanche front. This line becomes far more distinct at lower rotation rates, as does the interface between large and small particles near the wall. The striations also emerge below ∼10 rpm (Fig. 7e). The scale of striations grows with decreasing agitation, and structures depicted in Fig. 7(g) are larger and exhibit more intense segregation, than those in Fig. 7(e).

Similarly, unusual segregation is observed when the agitation direction is reversed to provide an acute blade pitch (Fig. 8). Large particle segregation is observed at high rotation rates in Fig. 8(a), but this is not particularly intense. However, at low shear, (Fig. 8b), more complex segregation emerges with a band of large particles accumulating midway between the agitator shaft and the walls. Striations are also visible, but in contrast to those for the obtuse blade pitch, form lines radiating from each of the four sink points between blades, far from the walls. Vertical segregation is shown in Fig. 8(c), in which large particles also overlay a thin layer of small particles. Close-up images of surface striations in Figs. 8(d)–(g) vary little with rotation rate.

Relations between segregation patterns and measured granular temperature gradients are not obvious. In Fig. 9, we plot mean velocity and granular temperature fields from PIV data, for high- and low-shear flows and for both blade pitches. Apart from the expected order of magnitude decrease in both velocities and temperatures at lower agitation, we see that relative granular temperature variations are different in all four cases. The obtuse blade pitch provides temperature maxima near the walls at both rotation rates, but for 50 rpm, this accompanied by a second temperature maxima near the shaft and a temperature well is thus observed midway between the walls and the shaft. This situation is reversed for the acute blade pitch, with a temperature maxima observed in this central location. The variations with blade pitch are complex, as shown in radial profiles

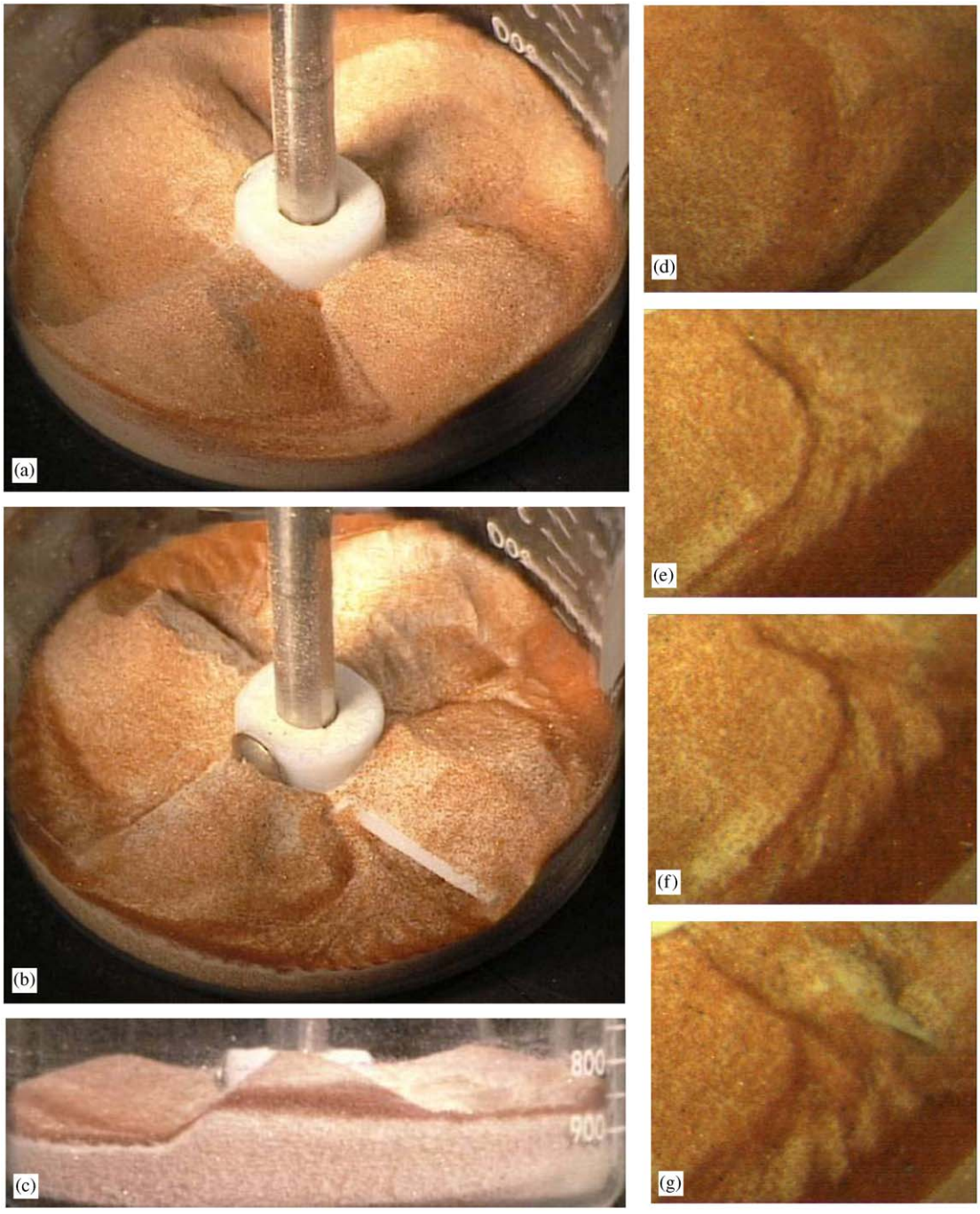



Fig. 7. Variation of segregation patterns with agitation rate for obtuse blade pitch, 125 and 450 μm glass spheres with counter-clockwise rotation. Top views shown for (a) 50 rpm, (b) 1 rpm. (c) View of vertical segregation taken at 1 rpm. Close-up views of striations developing in valleys behind blade for (d) 50 rpm, (e) 10 rpm, (f) 3 rpm, (g) 1 rpm.

for a wider range of agitation speeds in Fig. 10. The major effect of switching to the acute pitch is to increase radial velocities at the outer wall ($R \sim 45$ mm), signifying greater inwards particle motion (especially at higher rotation rates). Temperatures are lower than for the obtuse blade pitch and we evidently decrease particle fluctuation (and associated diffusional mixing) by switching to the acute configuration.

Comparison of observed segregation in Figs. 7 and 8 with temperature profiles achieves only a partial explanation using thermal diffusion arguments. While migration of large particles to regions of high granular temperature occurs at high rotation rates, no clear relation between temperature gradients and segregation exists at low shear. Kinetic theory predicts that gradients measured at low shear would

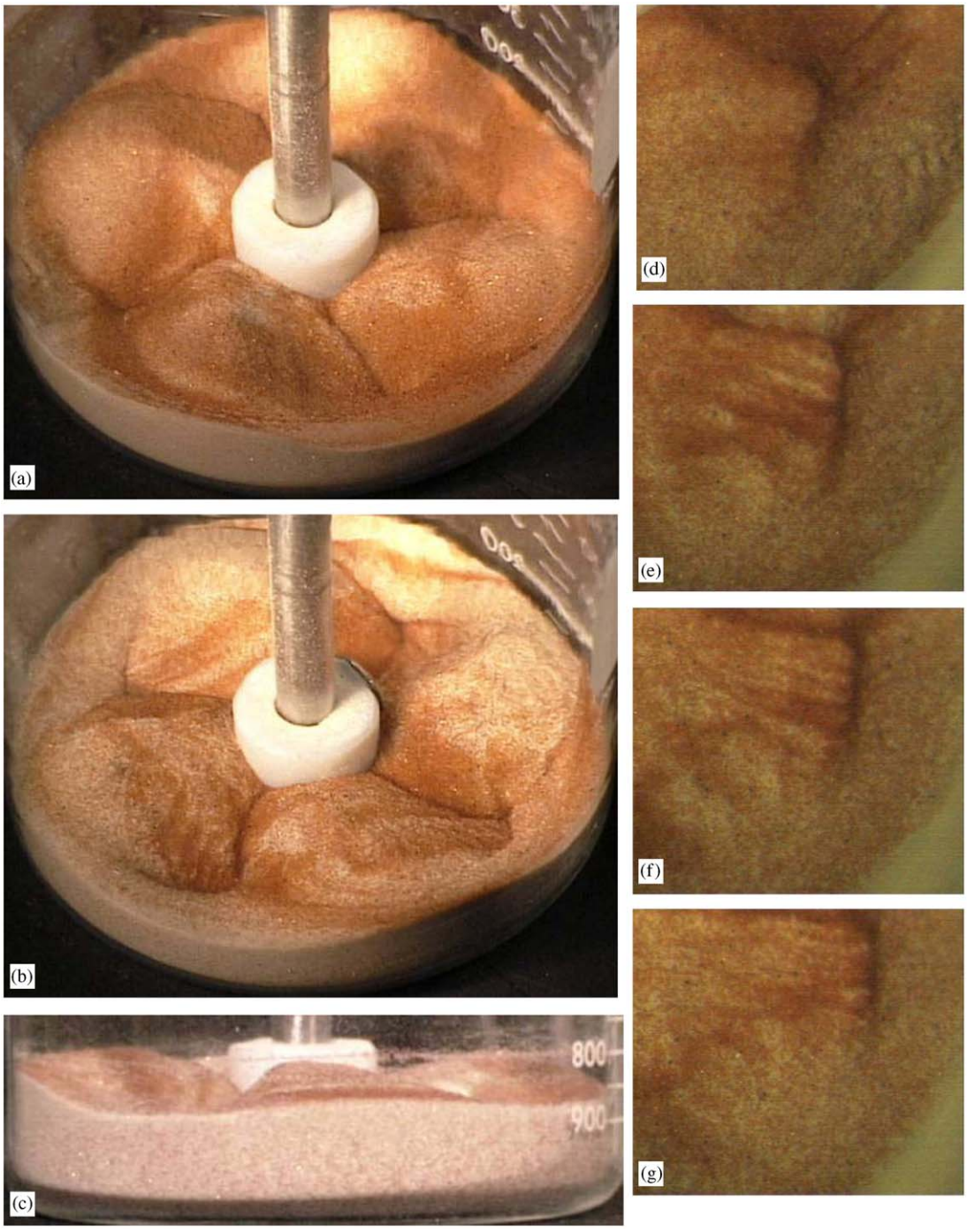



Fig. 8. Variation of segregation patterns with agitation rate for acute blade pitch, 125 and 450 μm glass spheres with counter-clockwise rotation. Top views shown for (a) 50 rpm, (b) 1 rpm. (c) View of vertical segregation taken at 1 rpm. Close-up views of striations developing in valleys behind blade for (d) 50 rpm, (e) 10 rpm, (f) 3 rpm, (g) 1 rpm.

drive large particles towards the central shaft. This is not the case in experiments. We also see little difference in granular temperatures between obtuse and acute orientations, yet these yield very different segregation patterns at low rotation rates. An alternative explanation for low-shear segregation is gleaned from inclined plane experiments. Segregation in such flows is well known, and is shown in Fig. 11(a), in which a binary mixture impinging upon an inclined plane, forming an unconfined heap, exhibits angle-of-repose segregation (Makse et al., 1997). Large particles have a lower angle of repose than the smaller (18° vs. 21°), and are thus able to flow when their free surface is at a lower incline.

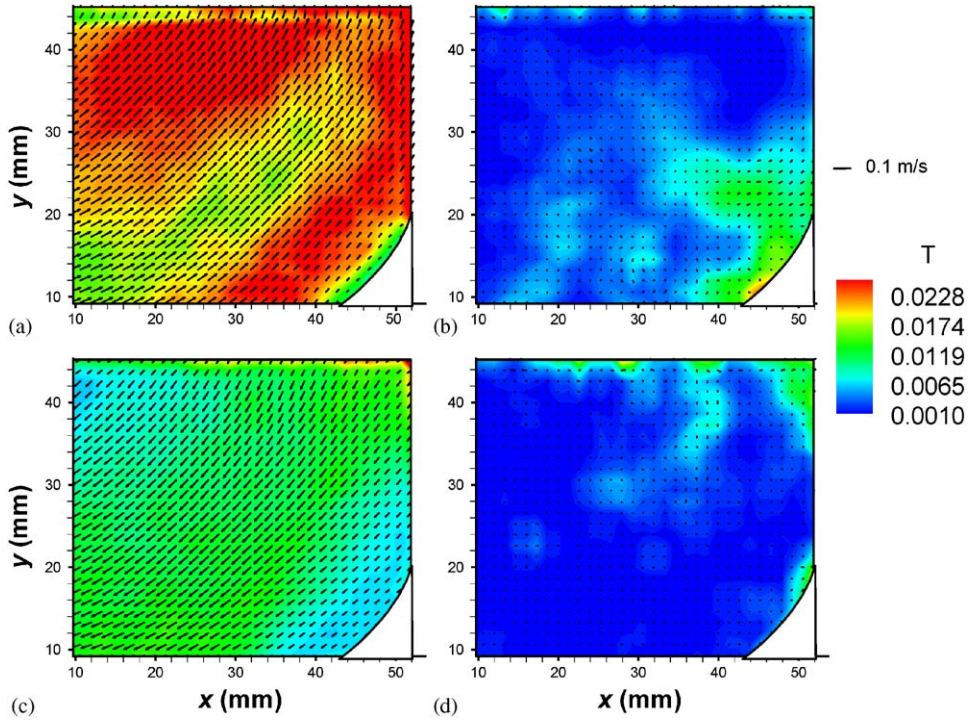



Fig. 9. Comparison of velocity and granular temperature fields, obtained from PIV of top views. For obtuse blade pitch: (a) high-shear operation at 50 rpm, (b) low-shear operation at 1 rpm. For acute blade pitch: (c) 50 rpm, and (d) 1 rpm.

Thus, in Fig. 11(a), the small particles form a steeper heap than the larger, which, in turn, are free to roll further down slope. They thus accumulate at the avalanche front.

The angle-of-repose segregation mechanism is applied to the current apparatus (under low-shear operation). We measure the angle of repose (i.e., the maximum incline at which a free surface remains static) for the large, rust particles as $26.8° \pm 1.4°$, and for the small, white particles as $31.6° \pm 3.5°$. Thus, in an avalanche, the larger particles are expected to congregate at the leading edge, and this appears consistent with observations of curved regions of large particles in Fig. 7(b). A schematic cross section of blade crest in Fig. 11(b) indicates how avalanches lead to segregation. The layering of large particles atop the smaller in Figs. 7(c) and 8(c) is also consistent with angle-of-repose segregation during avalanches via a process similar to that causing stratification in Hele–Shaw cells (Makse et al., 1997).

When this 2D process occurs in 3D flows, a mechanism of formation for the surface striations is conjectured. We previously observed that avalanche frequency increases with radial position. For a fixed rotation rate, we confirm this by generating power spectra from Fourier transforms of 1 s of azimuthal velocity data using *MATLAB*. Spectra for near-shaft and mid-steam locations are compared in Fig. 11(c). The strongest frequency signatures occur below 5 Hz, reflecting the passage of the blades, but we also see frequency features at 10–15 Hz. We propose that these are due to avalanches, and crude determinations of avalanche frequency using digital video footage supports this. The amplitude of power spectra peaks increases with distance from the shaft, consistent with our observations and PIV data (Fig. 5), in which the frequency of small-scale fluctuations also grows with radial position. This is not surprising, as more material is swept out by the blades as $R$ increases, and the number of avalanches should increase with the rate of material accumulation. Consequently, the striations caused by segregation in individual avalanche fronts do not extend over the entire blade length, and the scale of striations increases as agitation speed decreases, because segregated regions are disturbed less by subsequent avalanches. The location of segregation zones changes with blade pitch as the exact shape of the crest heap also changes with blade pitch, and avalanches fall in different directions. Finally, the formation of lines near sink points, depicted in Figs. 8(d)–(g), is proposed to be a consequence of convection segregation: large particles become trapped at the free surface as they are unable to penetrate narrow zones of submerging particles (Knight et al., 1993).

### 3.2.3. Mixing enhancement due to surface avalanching

While the patterns described reach steady-states in a matter of seconds after initiating agitation, it is important to quantify the kinetics of the initial transients for processes scaling. We find that the striations evident under low shear influence evolution of the flow structures. In Figs. 12(a) and (b), images of the surface of an initially left–right

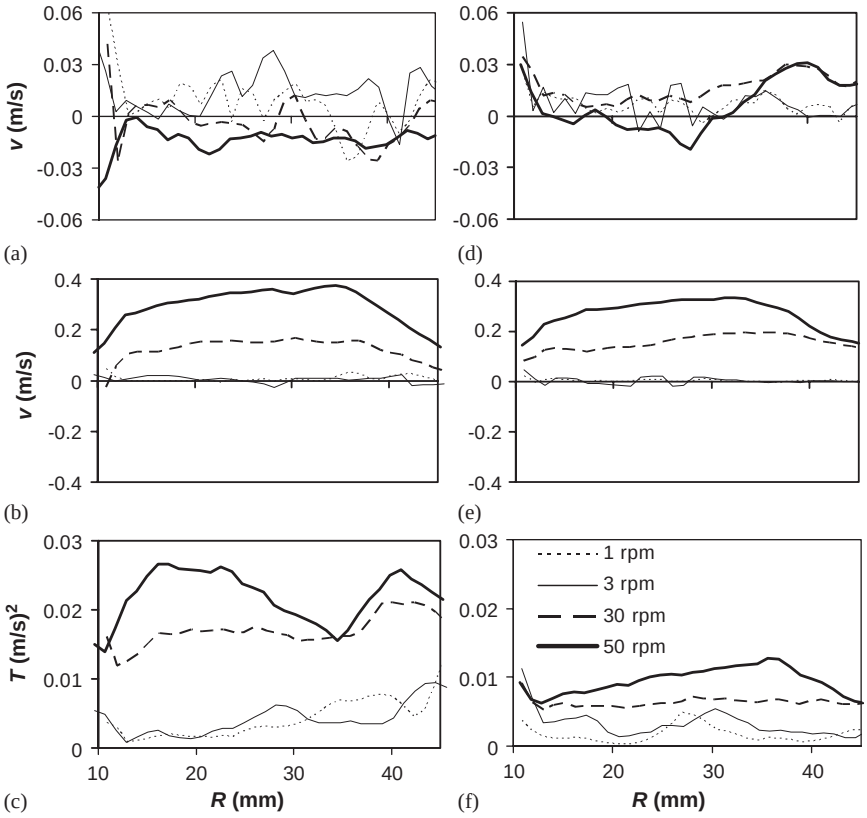



Fig. 10. Radial profiles from PIV for (a) radial velocity, (b) azimuthal velocity, and (c) granular temperature as a function of rotation rate for obtuse blade. Profiles for acute pitch: (d) radial velocity, (e) azimuthal velocity, and (f) granular temperature. (Heavy lines represent high-shear operations, light lines for low shear.)

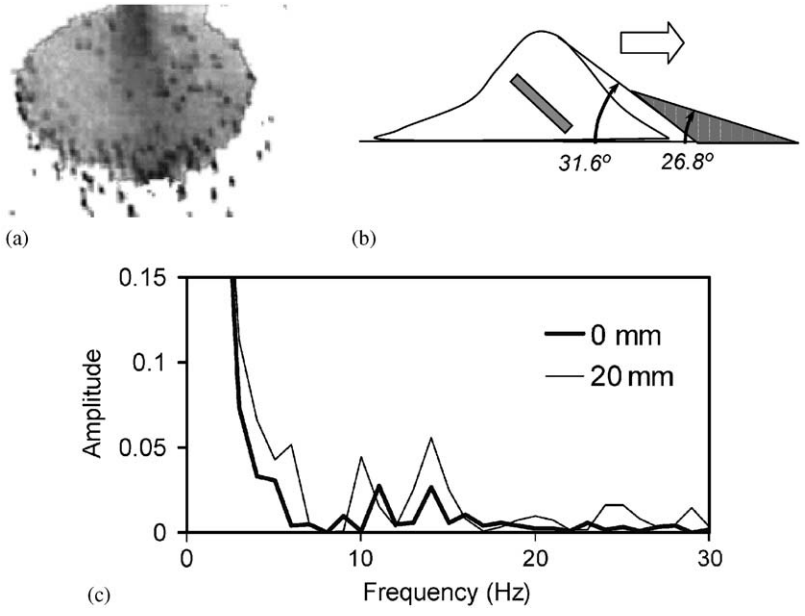



Fig. 11. Angle-of-repose segregation due to avalanches. (a) Granular mixture impinging on inclined surface forming a circular heap. Large, dark (4 mm particles) avalanche more readily than small, light (1.6 mm) particles. Large particles thereby segregate to the outskirts of a heap. (b) Sideview schematic of heap caused by passing blade. As material avalanches down free surface, angle-of-repose segregation drives larger particles to edges of flow. Arrow indicates direction of originating avalanche. (c) Power spectrum of surface velocity fluctuations for two radial positions. Blade rotation = 3 rpm. Amplitude of high-frequency avalanches increases with increasing distance from shaft.

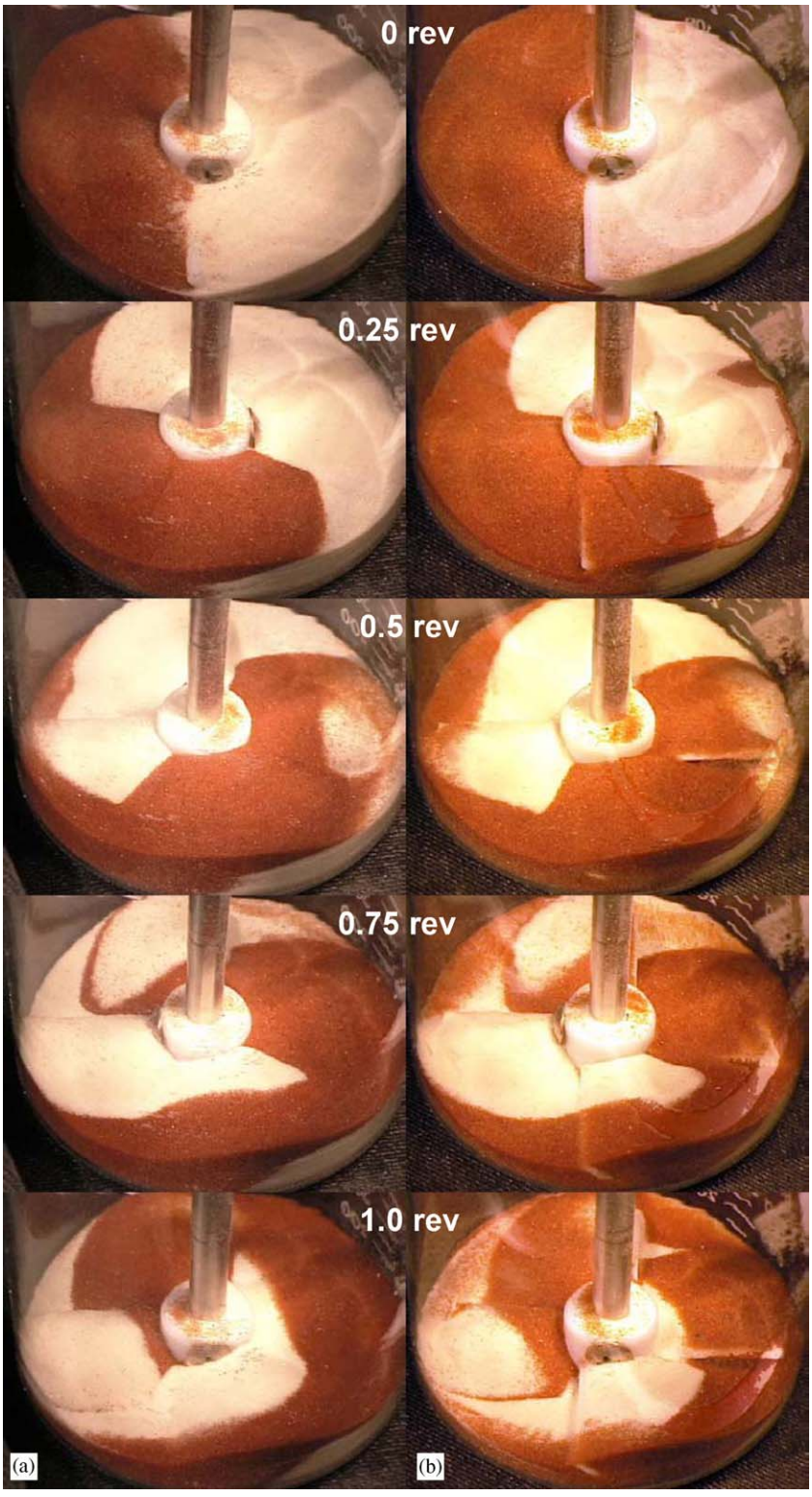



Fig. 12. Initial transient structures and interface evolution for initially left–right segregated bed. Images taken for increasing agitator revolutions for (a) high-shear operation at 30 rpm, and (b) low-shear operation at 3 rpm.

segregated bed are shown for increasing agitator revolutions, and images from high- and low-shear conditions are shown side by side. While bulk convective patterns evolve similarly with increasing blade motion (closely matching computer simulations of mixing in monodisperse flows (Zhou et al., 2004)), diffuse regions and striations emerge in the

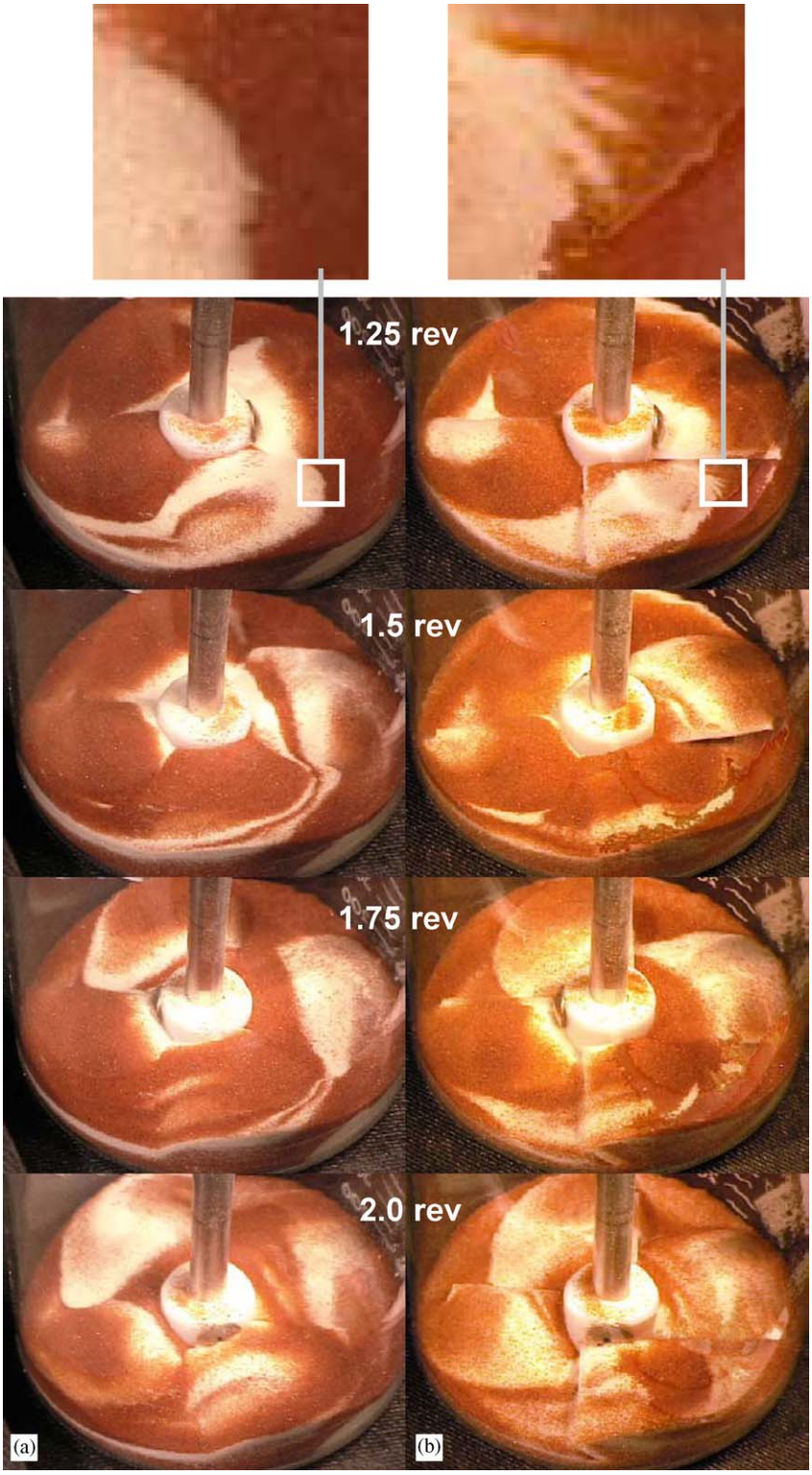



Fig. 12. (*Continued*)

low-shear regime that are absent at high shear. We find these features stretch the interface between the large and small particles considerably more that in the unstriated case, and we quantify this in Fig. 13. We use image analysis to highlight pixels in the surface images bordering both large and small particles, and counting these pixels provides a measurement of interface length (Shinbrot et al., 1999a,b). This length increases nearly exponentially in the low-shear

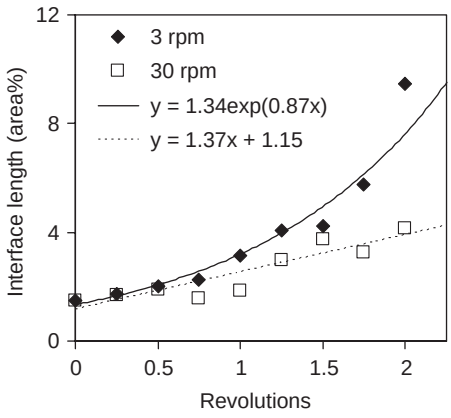



Fig. 13. Stretching of interface between initially left–right segregated granular bed. Interface length obtained by image analysis of digital video stills from number of pixels neighboring both rust and white particles, and shown as a percentage of the total number of pixels visible on the free surface. High-shear operation at 30 rpm (open symbols) exhibits linear stretching, while low-shear operation at 3 rpm (open symbols) appears to initially exhibit exponential stretching.

regime, and far more rapidly than the linear stretching observed at high shear. Exponential stretching is indicative of chaotic systems, but since the resolution of the image analysis does not extent to very diffuse, fully developed flows, we are unable to confirm true chaos is maintained over extended time periods. Nevertheless, because interface length indicates the degree of mixing between the two components of the mixture, we obtain the surprising result that mixing occurs more rapidly (with number of blade rotations) when the agitator speed is *lower*. This is a direct result of the avalanching and iterative reorientation, as also occurs in chaotically mixed tumbling blenders due to stick-slip motion at the walls (Shinbrot et al., 1999b). The degree of material avalanching in our case is found to be dependent on both agitation speed and location, but may just as easily depend on material properties. Indeed, we do not observe striations in mixtures of larger particles which are arguably subject to smaller cohesive effects, under any conditions (Figs. 6a and b). This may explain why segregation patterns in particle-dynamics simulations of cohesionless materials broadly agree with our observations at high shear, but provide no indication of the segregation we observe at low shear (Zhou et al., 2003). The avalanches thus appear to provide internal temporal forcing, similar to the role of viscoelasticity in inducing chaotic mixing in laminar 2D flows. Indeed, many of the lobe-like structures observed in the granular bed resemble the self-similar heterogeneous fluid structure in stirred tanks (Szalai et al., 2003).

## 4. Summary

Characterizing and understanding the mixing of powders is vital for prediction and optimization of the performances of many unit operations. The cylindrical bladed mixer configuration considered in this study is relevant to industrial applications ranging from homogeneity control to agglomeration/granulation processes but details of particle flow patterns in such mixers are limited, especially with regard to mixing mechanisms and segregation patterns.

In this paper, we used PIV technique to characterize 2D particle velocity fields of nearly monodisperse and polydisperse dry materials. We found that the bed free surface deforms, in accordance with blade position, causing particles to circulate between blades passes. Examination of instantaneous velocity fields revealed particle motion is downward on these slopes, indicating avalanching, and particles subsequently penetrate the bed. Segregation observed at high shear rates for polydisperse materials matches that observed in model geometries in the presence of granular temperature gradients. Conversely, drastically different segregation patterns were observed under low-shear conditions composed of lobe and striation structures. These are reminiscent of the stretching and folding features observed for liquid mixing in the laminar regime. We propose that this originates from internal temporal forcing caused by avalanches, and show that initial mixing rates are enhanced dramatically in a manner consistent with spontaneous chaotic granular mixing. These new features serve to highlight the limits of applicability of the rapid flow regime and current granular kinetic theory. However, we should note that these conclusions are based on results for one particular piece of equipment, and further work on different size units and different geometries are needed to develop more general conclusions.

While we only obtain 2D velocity fields, the resolution and instantaneous data obtained served as a complement to the fully 3D information from techniques such as PEPT. It follows from the above discussion that practical implications of the work include a greater level of detail in velocity fields and fluctuations than is typical for other experimental techniques, effectively enabling a new understanding of mixing mechanisms and potential segregation patterns. The latter directly impacts particulate mixture homogeneity, important in granulation or drying processes, in which granules of varying diameter coexist under agitation. Thus, improvements in product uniformity may be achieved through judicious choice of sampling locations or through process or equipment modifications. This study is meant to be the first step towards understanding granular flow in real processes, such as granulation and contact drying, where solid flow patterns, which impact the final product properties, are strongly affected by the presence of a second phase (liquid solvent). This will be the focus of future work.

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